IDENTICAL COMPLETE AXIS SETS DO NOT IMPLY AMBIENT COMMENSURABILITY OF KLEINIAN GROUPS

ARGUS OPEN RESEARCH PROJECT

Abstract. Question 5.1 of Bestvina's problem list asks whether two finitely generated Kleinian groups in $\mathrm{PSL}_2(\mathbb{C})$ sharing the same set of loxodromic axes must be commensurable. Following Xie and Jiang (2008), we record the elementary counterexample given by two cyclic loxodromic groups with multiplicatively independent translation lengths along the same geodesic.

1. The Elementary Counterexample

Let $\ell \subset \mathbb{H}^3$ be the vertical geodesic connecting 0 and ∞ in the upper half-space model, and define

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 1/3 \end{pmatrix} \in \mathrm{PSL}_2(\mathbb{C}).$$

Proposition 1.1. *The cyclic subgroups $G_1 = \langle A \rangle$ and $G_2 = \langle B \rangle$ are finitely generated discrete subgroups of $\mathrm{PSL}_2(\mathbb{C})$ whose nonidentity elements all have the same axis $\mathrm{Ax}(G_1) = \mathrm{Ax}(G_2) = \{\ell\}$, yet $G_1 \cap G_2 = \{I\}$ because $4^m = 9^n$ in $\mathbb{Q}^\times$ forces $m = n = 0$. Hence $[G_1 : G_1 \cap G_2] = [G_2 : G_1 \cap G_2] = \infty$, and G_1, G_2 are not commensurable in $\mathrm{PSL}_2(\mathbb{C})$.*

Remark 1.2. Both G_1 and G_2 are elementary and $\langle G_1, G_2 \rangle$ is indiscrete; the non-elementary Kleinian version holds by results of Long– Reid and Xie–Jiang.

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